Knowledge Graphs in Modern AI: MCP, Multi-Agent, and A2A

Block 1 Day 4

Semester 1, 2026–2027 — Teaching Week 3

Session Overview

Session objective: Connect graph-backed reasoning to MCP, multi-agent orchestration, and agent-to-agent handoff.
Timing target (min): 72
Topic anchors: knowledge graph; mcp; multi-agent; a2a; orchestration; neuro-symbolic
Padlet board: https://padlet.com/qmul/b3-d4-jbnj915ebdm59oqn

Exercises

Q1 Core Concept Check

Question type Concept → Working mode Pair → Expected time 9 min

Prompt

Match these four graph roles to a modern AI system and give one example for each: retrieval for RAG, long-term memory, explanation trace, and multimodal grounding.

Task details

- **Type:** concept
- **Mode:** pair
- **Time:** 9 min
- **Deliverable:** role-to-example mapping
- **Assessment focus:** connect graph roles to practical modern AI uses
- **Expected length:** 4 rows

How to submit

Post one pair Padlet comment as a four-row list or mini table.

Discussion in class

Which role would be easiest to remove, and what would break first?

Why this helps

Useful for short questions about what a knowledge graph adds to a modern AI pipeline.

References

- Russell & Norvig (2021) AIMA 4e, Ch.10, pp.314-343
- Poole & Mackworth (2023) 3e, Ch.16-17, pp.701-766
- doi:10.1016/j.cosrev.2026.100902

Q2 Structured Problem

Question type	Working mode	Expected time
Problem	Individual	11 min

Prompt

Put these Model Context Protocol (MCP) steps in a sensible order: user intent parse, capability discovery, tool selection, tool call, evidence return, response synthesis, audit log. Add one guardrail beside any two steps.

Task details

- **Type:** problem
- **Mode:** individual
- **Time:** 11 min
- **Deliverable:** ordered workflow
- **Assessment focus:** order an MCP request flow and attach guardrails
- **Expected length:** 7 ordered steps + 2 guardrails

How to submit

Post your own Padlet comment with the ordered list and two short guardrails.

Discussion in class

Which step is most dangerous if it is skipped?

Why this helps

Direct practice for workflow-ordering questions about protocol reasoning and tool use.

References

- Russell & Norvig (2021) AIMA 4e, Ch.10, pp.314-343
- Poole & Mackworth (2023) 3e, Ch.16-17, pp.701-766
- https://modelcontextprotocol.io/specification/

Q3 Structured Problem

Question type	Working mode	Expected time
Problem	Individual	12 min

Prompt

A timetable assistant uses MCP to query a room-change service. Explain the job of each of these four parts in that interaction: the client, the MCP server, the tool or resource being exposed, and the returned evidence.

Task details

- **Type:** problem
- **Mode:** individual
- **Time:** 12 min
- **Deliverable:** role explanation
- **Assessment focus:** explain the distinct roles inside a simple MCP interaction
- **Expected length:** 4 labelled bullets

How to submit

Post your own Padlet comment with four labelled bullets and one sentence on why this separation helps.

Discussion in class

Which role should own access control decisions?

Why this helps

Useful for questions that ask students to explain protocol components rather than only list them.

References

- Russell & Norvig (2021) AIMA 4e, Ch.10, pp.314-343
- Poole & Mackworth (2023) 3e, Ch.16-17, pp.701-766
- https://modelcontextprotocol.io/specification/

Q4 Collaborative Mini-Case

Question type
Case → **Working mode**
Group → **Expected time**
14 min

Prompt

Design a multi-agent workflow for cybersecurity triage using four roles: detector, correlator, responder, and governance. Include one Agent-to-Agent (A2A) handoff to an external threat-intelligence agent and one trust control before the handoff is accepted.

Task details

- **Type:** case
- **Mode:** group
- **Time:** 14 min
- **Deliverable:** orchestration design
- **Assessment focus:** design a multi-agent workflow with one A2A handoff and one governance control
- **Expected length:** 4 roles + 1 handoff + 1 control

How to submit

Post one group Padlet comment with the role order, the A2A handoff point, and one governance control.

Discussion in class

Which role should be allowed to trigger real action, and which should only advise?

Why this helps

Supports case questions on orchestration, handoff, and governance in distributed agent systems.

References

- Russell & Norvig (2021) AIMA 4e, Ch.10, pp.314-343
- Poole & Mackworth (2023) 3e, Ch.16-17, pp.701-766
- https://a2a-protocol.org/dev/

Q5 Exam-Style Constructed Response

Question type Exam → Working mode Pair → Expected time 14 min

Prompt

Explain when a knowledge-graph-enhanced pipeline is better than plain text retrieval for a knowledge assistant. Your answer should discuss one strong benefit and one operational risk.

Task details

- **Type:** exam
- **Mode:** pair
- **Time:** 14 min
- **Deliverable:** argument with trade-offs
- **Assessment focus:** explain when a KG-enhanced pipeline is better than plain text retrieval and what risk remains
- **Expected length:** 180-250 words

How to submit

Post one pair Padlet comment with a clear claim, supporting evidence, and one risk-control idea.

Discussion in class

Is the extra graph maintenance worth it for every assistant?

Why this helps

Matches exam answers that ask students to weigh architecture benefits against operational costs.

References

- Russell & Norvig (2021) AIMA 4e, Ch.10, pp.314-343
- Poole & Mackworth (2023) 3e, Ch.16-17, pp.701-766
- doi:10.1016/j.cosrev.2026.100902
- doi:10.1016/j.iswa.2025.200541

Q6 Stretch Extension

Question type Stretch → Working mode Group → Expected time 12 min

Prompt

Choose one neuro-symbolic pattern from the lecture - neural-to-symbolic, symbolic-to-neural, or joint constrained reasoning. Propose one concrete use case and explain why that pattern fits better than the other two.

Task details

- **Type:** stretch
- **Mode:** group

- **Time:** 12 min
- **Deliverable:** use-case proposal
- **Assessment focus:** choose a neuro-symbolic pattern and justify a realistic application
- **Expected length:** 4 bullets

How to submit

Post one group Padlet comment naming the pattern, the use case, and two reasons for the fit.

Discussion in class

Which pattern is easiest to explain to a non-technical manager?

Why this helps

Good extension for synthesis questions on combining symbolic structure with learned models.

References

- Russell & Norvig (2021) AIMA 4e, Ch.10, pp.314-343
- Poole & Mackworth (2023) 3e, Ch.16-17, pp.701-766
- doi:10.1109/TNNLS.2024.3420218
- doi:10.1016/j.iswa.2025.200541

Submission Instructions

Use the seeded Padlet posts for Q1–Q6. Q2 and Q3 are individual. Q1 and Q5 are pair tasks. Q4 and Q6 should be one joint group comment. Start each comment with your names or group label.

Whole-Class Debrief Prompts

- Which part of the MCP flow most clearly needs a guardrail?
- What makes an A2A handoff trustworthy instead of dangerous?
- When is a graph-enhanced pipeline worth the extra maintenance cost?

Exam Transfer Note (Day-Level)

Maps to exam questions on graph roles in modern AI, MCP flow, protocol roles, multi-agent orchestration, A2A handoff, and architecture trade-offs.